Bio-Waste Fermentation & Yield Calculator

Minimum Viable Product (MVP) — Project Brief

Strategic Alignment: SDG 7 (Clean Energy) & SDG 12 (Responsible Consumption)
Target Context: Sustainability Transformation in Rural Areas (VISTRA)

Executive Summary

In regions with intensive agriculture, such as the Münsterland and Vechta areas, the management of organic agricultural waste presents both a logistical challenge and a significant ecological opportunity. This Minimum Viable Product (MVP) is a geospatial decision-support tool designed to quantify the latent energy potential within local biomass. By transforming complex biochemical methane potential (BMP) calculations into an accessible digital interface, the application empowers rural municipalities, local farmers, and researchers to instantly calculate the biogas and electrical equivalents of their available organic waste.

Core Features & Functionality

The system provides an intuitive workflow to bridge raw agricultural data with actionable sustainability metrics:

- **Algorithmic Yield Estimation:** Users select specific organic feedstocks (e.g., swine manure, mixed food waste, wheat straw) and input tonnage. The engine instantly calculates estimated volumetric biogas yield (m^3) and electrical conversion equivalents (kWh) based on baseline biochemical factors.
- **Geospatial Resource Mapping:** Integrated interactive mapping allows users to drop a pin on the exact location of the biomass. Reverse-geocoding automatically translates these coordinates into human-readable municipal addresses, enabling spatial planning for regional bio-energy logistics.
- **Bilingual Accessibility:** The interface features seamless, session-based toggling between English and German, ensuring the tool is immediately usable by both international researchers and local agricultural stakeholders.
- **Standardized Institutional Reporting:** The application generates dynamic, localized PDF reports containing the user's input data, estimated energy yields, and verified geographical coordinates, ready for submission to agricultural boards or funding bodies.

For institutes focused on spatial analysis and rural transformation, this MVP serves as a functional prototype demonstrating how software engineering can reduce "data friction" in renewable energy transitions, directly advancing the United Nations Sustainable Development Goals.

Technical Architecture

The system is built on a lightweight, scalable Python backend utilizing the Django framework. The frontend leverages responsive design principles alongside the Leaflet library and OpenStreetMap APIs for zero-overhead geospatial processing. Document generation is handled server-side, ensuring consistent, print-ready institutional reports independent of the user's local hardware or browser configuration.